

Part A Chapter 1

Introduction to Modelling and Simulation¹

1 What are models

Modelling is an important activity in many different areas. However, although the same term “modelling” is used, the actual task and the result of the activity can differ enormously. For example, property developers often build *scale models* to illustrate how a future building or a future neighbourhood will look like, economists build *economical models* to explain and predict economical phenomena, while software engineers design *conceptual models* of computer programs and data structures.

In all these different examples a model is *a representation of an object, system, or idea in some form other than that of the entity itself* (Shannon, 1975). Such a representation describes the most important elements and their relations. However, it is almost never possible to describe all these aspects completely and unambiguously, e.g. because the exact relation between elements in reality is not known, or because not all factors that influence an element are known. Therefore, the definition of Shannon emphasizes the fact that a model is different from what is modelled in reality. Although this might seem obvious, it has a number of important consequences:

1. a model is a simplification, not a complete representation of the reality;
2. a model is based on several choices and assumptions of the creator of the model;
3. characteristics of the model are not necessarily characteristics of reality.

These consequences should be taken into account when working with models, as they imply that some type of conclusions can be drawn from a model and some can not.

In the definition given by Shannon it is already made explicit that several things that can be modelled. Modelling is not just about physical objects, but also non-tangible things such as theories, imaginary concepts or interaction between objects can be modelled. An important



A planetarium is a model of the solar system.

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distinction is the difference between *static* and *dynamic* models. Static models describe the structural aspects of a system and provides a time-independent view of the system. Dynamic models account for the behaviour of a system and describe the interactions between aspects in the model; they provide a time-dependend view. The models on which we focus in this course are dynamic models. We will therefore generally use the term *process* for the subject of the model.

The general definition of a model given above ignores its usage. The definition in the “American Heritage Dictionary” refers to this aspect as well: *a model is a schematic description of a system, theory, or phenomenon that accounts for its known or inferred properties and may be used for further study of its characteristics*. According to this definition a possible goal of a model is to study the behaviour and characteristics of the part of the reality that is modelled. Studying the behaviour or characteristics could take several forms. For example, the model can be used to understand the interaction between elements in the reality, to predict what would happen in the future, to gain insight in the effects of changes in the environment, or to verify that specific unwanted situations will never occur.

There may be several possible reasons for studying a model instead of studying the process itself. One possible reason is that the process does not yet exist. For example, models of airplanes are used to study their aerodynamic features before the planes are actually built. Another reason can be that the process itself cannot be easily studied directly. For example, think about cognitive processes: it is not easy to measure things that go on in a human mind, while one can analyze relatively easily how a model of it behaves.² One other common reason for studying models instead of the process in reality is prediction: one would like to see what happens in the future. Economical and weather models are an example of models that are often used for prediction. Finally, experimenting using a model can be cheaper and more feasible than experimenting with real processes.

2 Simulation

In order to study a model, it is often useful if the model can imitate what would happen with the process over time. This process is called “simulation”, and models that can be used to simulate behaviour over time are called “dynamical models”. The result of a simulation is a sequence of states of the model at

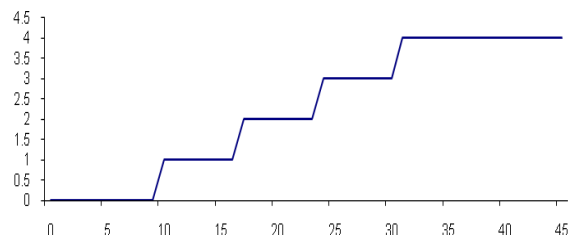


Figure 1. A graphical representation of the value of a parameter over time.

² Because a model is a simplification and based on assumptions, this is not by definition the same as what actually happens in the mind.

subsequent time points, which is called a “trace”, or “simulation trace”, or sometimes “simulation run”. For example, a simulation trace resulting from a model of a solar system can be a list of the locations of the planets at all days in a specific year.

Quite often graphical representations are used to visualize a simulation trace. Figure 1 shows a simple graph that visualizes the value of some variable in a model over time. The horizontal axis represents the time points, the vertical axis the values; note that the units are not specified in this figure. Another common

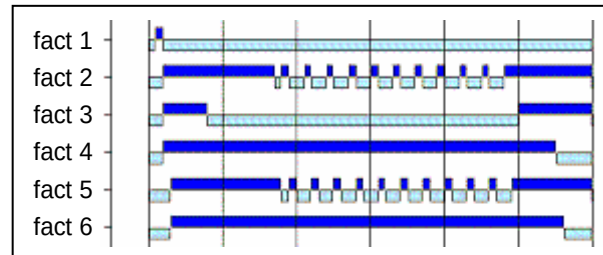


Figure 2. Example visualization of a state trace.

way to visualize a trace is shown in Figure 2. This picture lists 6 facts at the vertical axis and shows when they are true or false. The horizontal axis represents the time, again: the first time point is at the left side of the picture, the last time point is at the right. If a fact is true at a time interval there is a dark (blue) bar above the line, while there is a light (blue) bar below the line when the fact is false.

3 Modelling and simulation cycle

Modelling is a creative process. There are many choices that can be made, resulting in different models. Nevertheless, some guidelines can be given to structure the modelling process. A book about simulation formulates the essence of this process as follows: “The art of modelling is enhanced by an ability to abstract the essential features of a problem, to select and modify basic assumptions that characterize the system, and then to enrich and elaborate the model until a useful approximation results” (Banks and Carson, 1984).

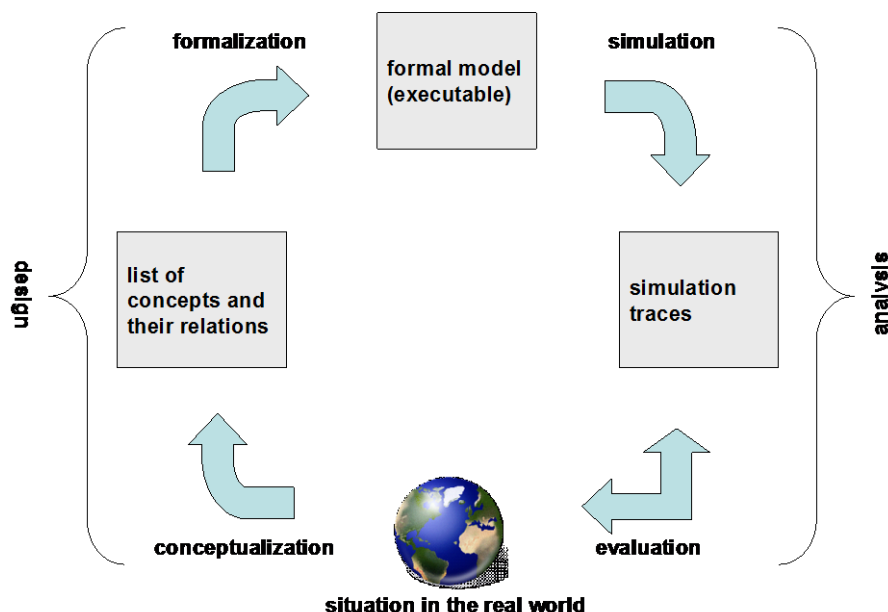


Figure 3. General process model for modelling

Figure 3 depicts a general process model for designing and analyzing models, the modelling and simulation cycle. There are two main phases: model *design*, in which the model is built, and model *analysis*, in which the model and its behaviour is studied.

The cycle consists of four activities:

1. **Conceptualization**: determining the main aspects and their relations;
2. **Formalization**: specifying the detailed model;
3. **Simulation**: performing experiments with the model to generate traces;
4. **Evaluation**: verifying whether the model behaves as expected.

If the evaluation reveals that the model is not correct, the process starts again with a new conceptualization phase. The next two sections of this chapter discuss in detail the activities in the design and analysis phase, respectively.

4 Designing a model

In this section, a methodology for building models is presented that is used to structure all subsequent chapters. However, it should be clear that steps can be taken in different orders and different paths can be followed. Designing a model consists of two activities: conceptualization and formalization.

4.1 Conceptualization

Within the conceptualization phase three types of tasks can be distinguished. These tasks may be executed in a *iterative* manner, as the insights that are gained in one task might influence choices in other tasks. It might be necessary to go back and make other choices.

Characterising the process to be modelled. First it has to be decided what process in reality the model should address. What is part of the process to be modelled and what falls outside the scope of the model? It can help to define beforehand the questions that you want to answer with your model. For example, for the definition of an economical model, it can be stated that the models should be able to answer the question what the effect is of increased interest rates on the employment in 5 years from now. It can also help to define the main *characteristic patterns* of the process that you want the model to exhibit. These patterns are the most important relations between elements in the process. For example, in a model that describes the effect of eating and physical activity on weight, you would like to see that an increase in the physical activity will lead to decrease in weight.

(Identification of sub-processes). For larger processes, it might help to divide the process into several sub-processes, for each of which you then follow the next steps. You partition the overall process into coherent sub-processes, and identify the interactions between them. A sketch could show the relation with the overall process, see for example Figure 4.

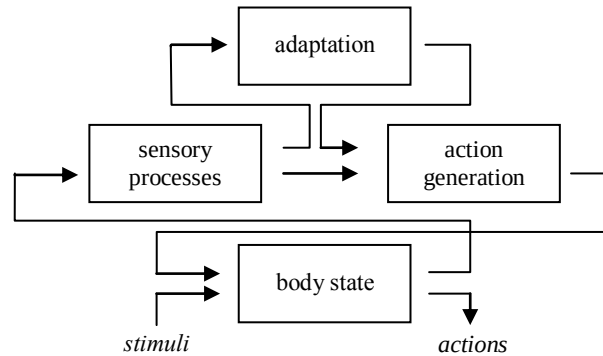


Figure 4. Example of a diagram with processes.

Concept identification. The next task is the identification of the relevant concepts in a process. A concept can be an object in the real world, but it can also be an event that is important in the process that is modelled. Which concepts are important completely depends on the process that is modelled. This is thus a creative activity, resulting in a list of concepts.

Relationships between concepts: A subsequent task is the specification of the relationships between the concepts that have been identified. In this stage, this only means that is specified whether a concept influences another concept, but not yet what the influence actually is. It should be possible to classify each relationship under one of the sub-processes identified before. The result of this activity can be a list of statements in the form of “A affects B”, where it is possible that concepts are influenced by multiple concepts, or that a concept influence multiple concepts. It is also an option to represent these statements as a graph. To do so, each concept should be depicted as a node (circle)

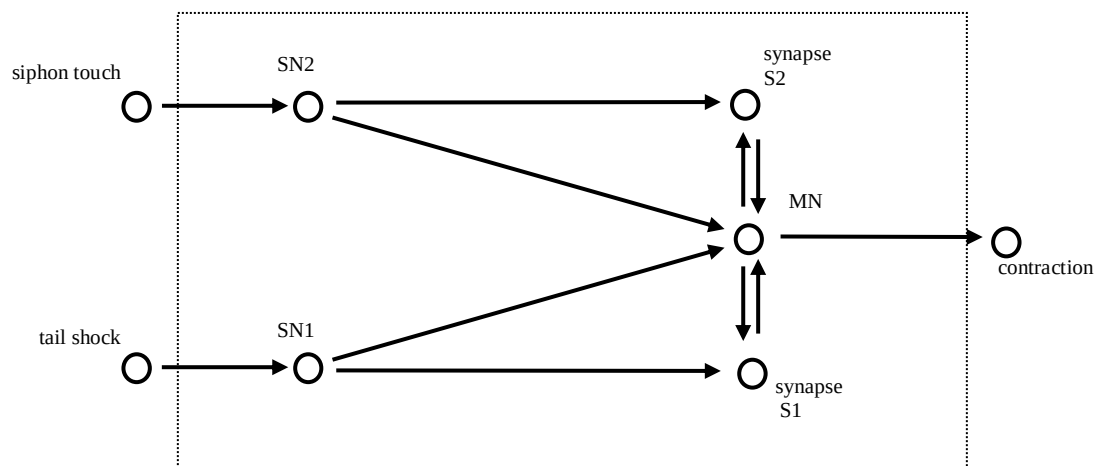


Figure 5. Diagram showing relationships between concepts

and the relationships as edges (arrows) between the nodes. For an example, see Figure 5 (taken from Chapter 4).

4.2 Formalization

In the formalization activity, the details for the abstract concepts and relations specified in the conceptualization phase will be filled in. This has to be done for both the concepts and the relations. The formalization depends on the representation mechanism that will be used for the model. Two representations will be discussed: numerical representation, and logical representation.

Numerical representation

In a numerical representation (which will be used in the chapters in Part A), concepts are represented by variables with a value. Relations are rules that specify how the value of one variable can be calculated from the values of other variables.

Formalization of concepts in a numerical representation is often quite straightforward. For each concept, variables are defined which can have certain values, often “0 or 1”, “integer numbers” or “real numbers”. For example, if the concept “temperature” is defined, a formalisation could be a variable “*Temp*” which can have all real values. Similarly, the concept “button pressed” could be formalized by a variable “*BP*” which could have the values “0 or 1”. Note that a “0 or 1” concept also can be used to represent a fact that is true or false, which is already a kind of logical representation.

The formalization of the relations requires a bit deeper thought. For each of the relations, it should be determined how the concepts exactly influence each other. This can also take several forms, depending on the values that a variable can have, and on the time delay related to the effect. A few are given below.

- A concept that affects another concept in a next time step can be specified as:

$$B2(t + \Delta t) = B1(t)$$

In this formula, ‘*t*’ is a variable representing the time, Δt (delta *t*) stands for a change in the time variable. Thus, the formula should be read as: “at the next time point *B2* will have the value of *B1* at the current time point”.

- If a concept depends on two other concepts the following can be specified:

$$B3(t + \Delta t) = B1(t) * (1 - B2(t))$$

This reads as: “at the next time point *B3* will have the value of *B1* at the current time point multiplied by 1 minus the value of *B2* at the current time point”.

- Sometimes values of concepts at the same time point can be related. For example,

$$R1(t) = \gamma * R2(t)$$

denotes that the value of $R1$ at a time point t is the value of $R2$ at the same time point multiplied by γ .

The following chapters will contain more examples.

Logical representation

In a logical representation (which will be used in the chapters in Part B), concepts and their properties are represented by statements that can be true or false. For example, a statement could be `medicine_taken(hiv_slows)`, which could represent the concept that it is true that the medicine “hiv slows” has been taken.

Relations between concepts are rules that specify statements such that when at some time point they are true then another statement will be true at another (next) time point. For example, $A \rightarrow B$ says that when A is true then B will be true. These rules can combine statements in several ways. A few often used combinations are listed below (note that for simplicity characters like A , B , etcetera, are used instead of statements of the form “`medicine_taken(hiv_slows)`”).

- $A \wedge B \rightarrow C$: if A is true and B is true, then C will be true;
- $A \vee B \rightarrow C$: if A is true, B is true, or both are true, then C will be true;
- $A(X) \rightarrow B(X)$: if A is true for a specific value of X , then B will be true for that value of X ;
- $\text{not}(A) \rightarrow B$: if A is not true, then B will be true;
- $A(X) \wedge X > 52 \rightarrow B$: if A is true for a specific value of X , which is larger than 52, then B will be true.

The chapters in Part B will contain more examples.

5 Analysis of and with a model

Although a model is technically finished when we have a formal representation of the concepts and their relations, the formalization is not the final step of model development. It is still necessary to check whether the model is correctly built and whether it represents the reality in an adequate way. This is called *validation*.

Validation is done by setting up and running simulation experiments and evaluating the resulting traces by comparing them to the expected outcome. This is shown in the modelling and simulation cycle described in Section 3.

However, it should be noted that the same steps (running experiments and evaluating the results) can also be used for other purposes than model validation. When we want to use the model to *study the process* that we have modelled, we also setup experiments and evaluate the outcome. For example, we could use a model of bacteria growth to investigate the effect of

changes in the available amount of food. Or, we could use a model of criminal behaviour to test whether a presumed relation between the number of policemen in a city and the number of burglaries is a consequence of a specific theory.

Because of these two different use cases for simulation and evaluation, we use the more generic term *analysis* for the two tasks at the right side of the modelling and simulation cycle. This can refer to *validation* when the analysis is used to check whether the model exhibits the expected behaviour (e.g. as formulated in the characteristic patterns) and to *studying the process* when the analysis is used to answer research questions about the modelled process. The steps below can thus be used for both analysis *of* the model and analysis *with* the model.

5.1 Simulation experiments

Dependent on the representation of the model and the software tools used, it is possible to simulate the model and generate a simulation trace. To perform a simulation experiment a number of steps have to be taken. First it is formulated what question or what pattern is to be addressed by the experiment, for example, one of the characteristic patterns identified in Section 4.1 in case of validation, or a research question in case the model is used to study the process. As a next step, often several choices can be made with respect to the initial values of variables, the values of the parameters, and values that are added by interaction during the process. Setting up a simulation experiment requires a structured plan (a *scenario*) on which values are chosen for these initial values, parameter values. Moreover, different scenarios can be used in order to compare the outcomes for different circumstances: changing the values in a number of different simulations. For a thorough evaluation, one often tries to simulate a number of characteristic scenarios that could happen in reality.

Some models contain a stochastic element, i.e. parameters for which the values are determined by a probability. With this kind of models, running the same model several times results in possibly different traces, which all reflect a different scenario for the same model.

5.2 Evaluation

Evaluation consists of comparing the simulation results with expectations. These expectations can be based on the reality (in case of validation) or on hypotheses (in case of studying the process). The expectations can be formulated as *properties* for which we then can check whether they hold. With the right software it is possible to formulate properties in a specific formal language and check automatically whether they hold in the model or the simulation traces.

A distinction can be made between two types of properties:

1. Quantitative properties: statements about numerical characteristics of a model. Examples of quantitative properties are: “the unemployment rate never reaches 10 percent”, or “the liquid boils within 4 minutes”, etc.
2. Qualitative properties: statements about non-numerical relations or characteristics of the model. Examples of qualitative properties are: “eventually state property X will be true”, “event A always occurs before event B”, “parameter X is always decreasing”, or “the simulation always reaches an equilibrium”, etc.

There are two fundamentally different ways to verify whether a property holds. First, one could check whether a property holds for all traces that have been generated based on the scenario's that were specified. This is a matter of checking for each of the generated traces (either manually or automatically) if the formulated characteristic holds, e.g. whether the value of some parameter is always below some threshold. However, when this has been proven, it doesn't mean that it is necessarily true in all possible situations. It could just happen that verified traces accidentally didn't contain a violation of the property. The other approach for verification looks at the logical characteristics of a model, and answers the question whether a property is *necessarily* true. The answer is giving by a mathematical or logical proof, in most cases by a computer program and not manually. This specific type of verification is called “model checking” in computer science. Note that this meaning is much more specific than “verifying a model” as discussed in this section.

6 Modelling human processes

Simulations are not only used to evaluate a model, but can also be used to understand the behaviour of the process or to make predictions. One can study the behaviour of the process by looking at a number of simulation traces under different circumstances. This could reveal some patterns, e.g. that the value of a parameter of the model is stepwise increasing (as was the case in Figure 1), or that some event always happens after another event. Also, one could make changes in the model to see the effect of such a change in the behaviour of the model. For example, when different initial values of the simulation are chosen, or when specific relations between components in the model are changed, a different outcome will be obtained. This can be useful in a prediction task; for example, in an economical model, one could check what the effect is of the interest rate on the employment rate in a country.

In this course the focus is on modelling of human processes. Such models can be used in different manners. One possibility is to design hardware and/or software systems that in their behaviour have certain aspects in common with humans. For example, they may have some forms of intelligence, or show some types of emotions or even consciousness. Also virtual characters in games or virtual stories fall in this category. Another possible use of models of

human processes is in software and/or hardware that interacts with a human and tries to understand this human's states and processes and responds in an intelligent manner: ambient intelligence. An example of this is a device that monitors elderly persons.

The course is structured according to three parts. In Part A the focus is on physiological processes and how they relate to cognitive and decision making processes. Part B focuses on a psychological context: cognitive processes and how they generate behaviour of individuals. Finally, Part C addresses how in a social context individual behaviours lead to collective patterns.

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